

# Introduction

Microplastics are an increasing environmental pollutant in freshwater and marine systems, and everyday products such as **nylon tea bags** are prone to releasing large quantities of these particles. A single tea bag can release approximately **11.6 billion microplastic particles** (Hernandez et al., 2019), many of which have been found in **human feces, blood, and placentas** (Schwabl et al., 2019; Leslie et al., 2022; Ragusa et al., 2020). Once released into aquatic ecosystems, microplastics can be ingested by organisms, potentially **disrupting cellular processes and overall health**.

By focusing on microplastics released from a real-world consumer product, this research addresses ecological and human health concerns. Changes in planarian regeneration can indicate cellular stress caused by environmental toxins and provide insight into how microplastics may impact **stem cell function and tissue repair** in humans, offering a way to study the potential risks of everyday plastic exposure.

## Why Planarians:

They regenerate using **neoblast stem cells**, which share key genetic and molecular pathways with **human stem cells**, meaning processes controlling growth, differentiation, and repair are similar between species.



Figure 1. Image showing microplastic size relative to human finger

[https://images.fastcompany.com/image/upload/fit\\_webp\\_q\\_auto,c\\_fit/wp-content/uploads/2020/08/poster-microplastics-organs-research.jpg](https://images.fastcompany.com/image/upload/fit_webp_q_auto,c_fit/wp-content/uploads/2020/08/poster-microplastics-organs-research.jpg)

# Objectives

## Purpose:

The purpose of this study is to investigate how exposure to microplastics released from nylon tea bags affects the **regenerative abilities** of *Dugesia tigrina*.

## Hypothesis:

If *Dugesia tigrina* are exposed to microplastics from nylon tea bags, then their regeneration after being cut will be significantly slower compared to planaria exposed only to spring water, because microplastics interfere with stem cell function and cellular processes.

## Variables:

The independent variable is the steep time of the tea bags and the dependent variable is the amount of regeneration in planarians.

## Control:

Planarians in the control group will be exposed to clean, microplastic-free spring water.

## Constants:

All planarians are the same species and similar in size and age. The water source and volume, environmental temperature, sunlight, feeding schedule, container type, and imaging settings are the same.

# The Effects of Microplastic Exposure from Nylon Tea Bags on the Regenerative Abilities of *Dugesia tigrina*

# Methodology

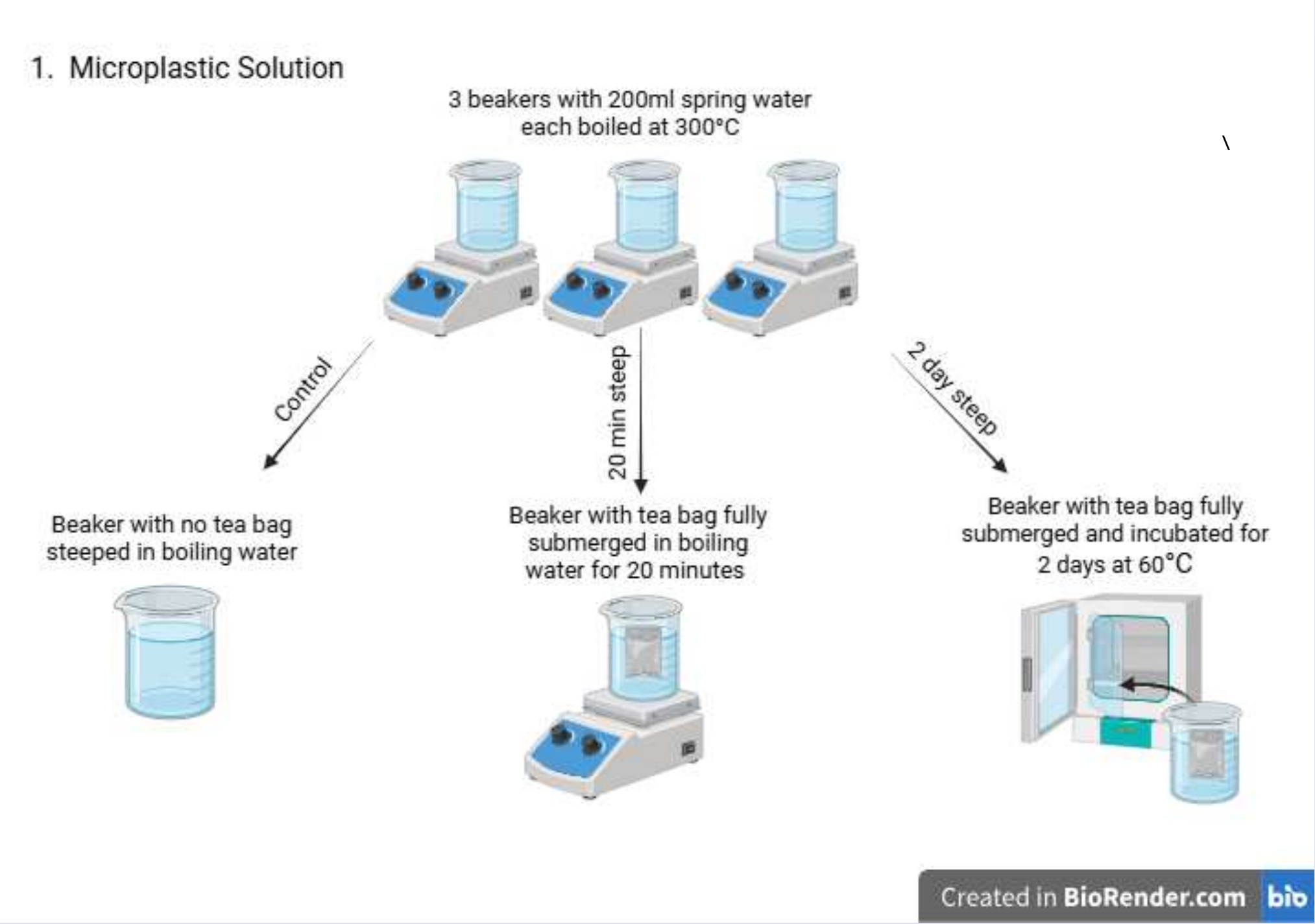


Figure 2. Illustration of first part of Experimental Design.

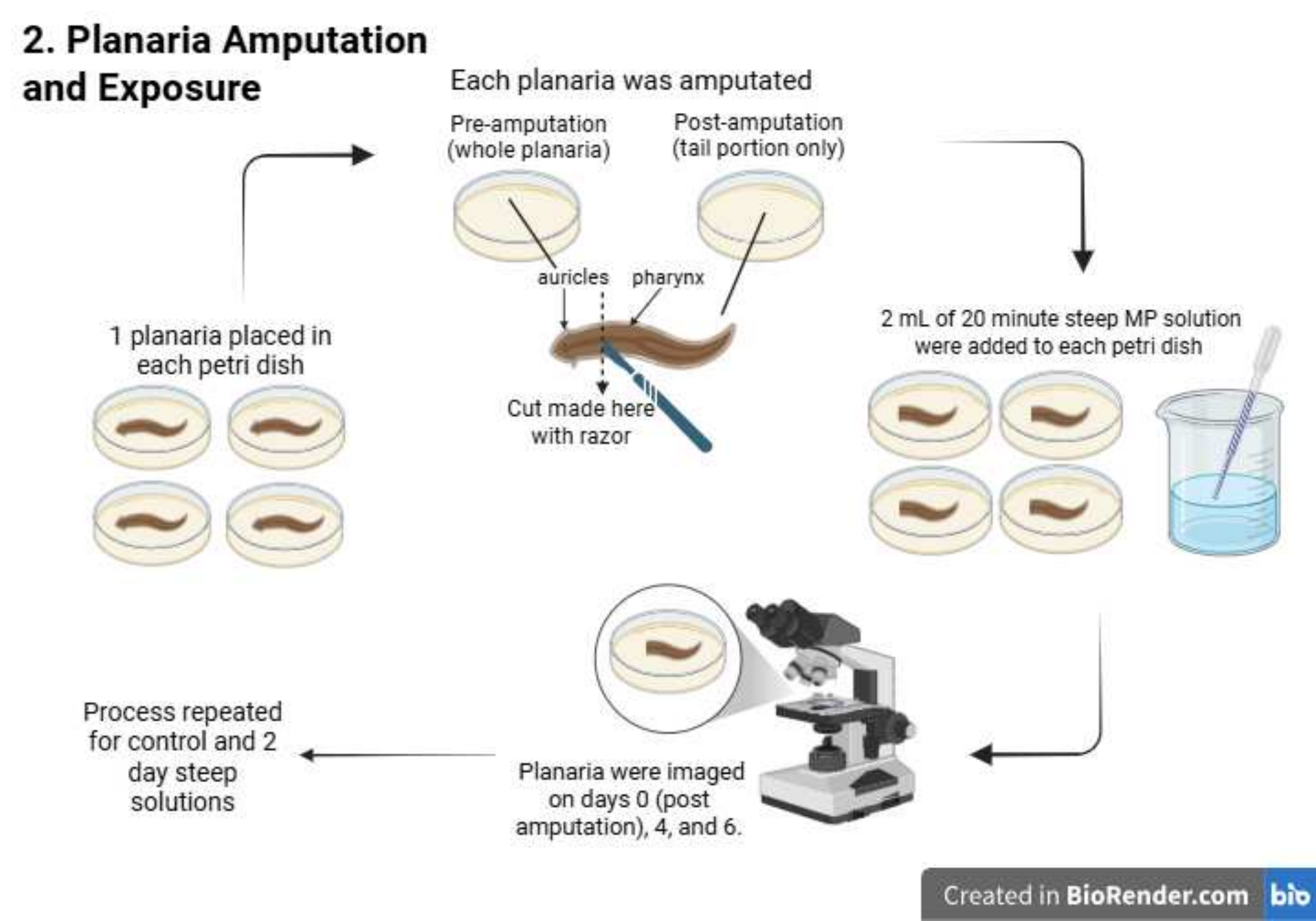


Figure 3. Illustration of second part of Experimental Design.

| Control | P1 (mm) | P2 (mm) | P3 (mm) | P4 (mm) |
|---------|---------|---------|---------|---------|
| Day 0   | 1.932   | 0.889   | 1.787   | 0.982   |
| Day 4   | 2.336   | 1.98    | 2.665   | 1.398   |
| Day 6   | 8.669   | 10.251  | 8.416   | 9.518   |

| 20-min | P1 (mm) | P2 (mm) | P3 (mm) | P4 (mm) |
|--------|---------|---------|---------|---------|
| Day 0  | 8.046   | 7.907   | 2.985   | 5.548   |
| Day 4  | 8.734   | 8.477   | 5.693   | 7.822   |
| Day 6  | 9.183   | 9.652   | 6.038   | 7.964   |

| 2 Hours | P1 (mm) | P2 (mm) | P3 (mm) | P4 (mm) |
|---------|---------|---------|---------|---------|
| Day 0   | 3.254   | 2.409   | 2.152   | 2.438   |
| Day 4   | 3.684   | 2.592   | 2.777   | 2.665   |
| Day 6   | 3.859   | 3.302   | 2.786   | 2.732   |

Figure 6. The differences in lengths from day 0 to day 6 decrease dramatically as the exposure time increases. This is consistent across all 4 trial groups.

## Making the Microplastic Solution

- Nylon tea bags were steeped in 200 mL of boiling spring water for different durations (control, 20 minutes, and 2 days) to produce varying concentrations of microplastics.

## Amputation and Exposure of Planarian

- Dugesia tigrina* were cut between the eyes and pharynx to separate head and tail fragments.
- Tail fragments were placed into individual petri dishes containing 20ml of control or treatment solution.
- Petri dishes were labeled by group, trial number, and planarian ID
- Regeneration was observed over an 6-day period.
- Images were captured using a dissecting microscope and iPhone camera.
- Regeneration length (mm) was measured from images using ImageJ software.
- Regeneration growth recorded and compared across all treatment groups



Figure 4. Image taken of planarian on day 4 post-amputation

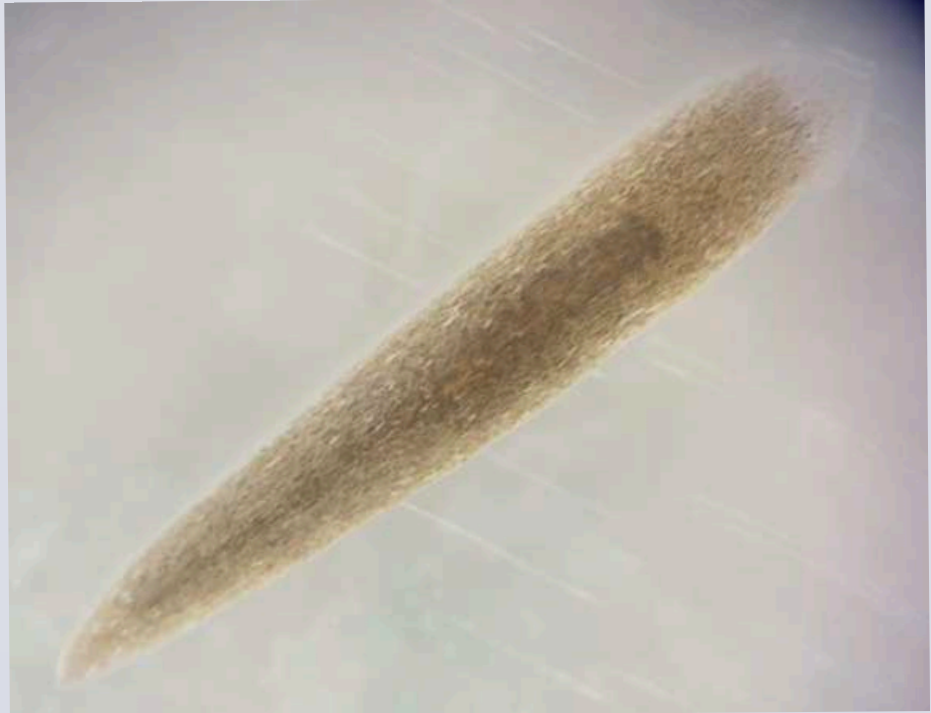


Figure 5. Image taken of planarian on day 6 post-amputation

# Results

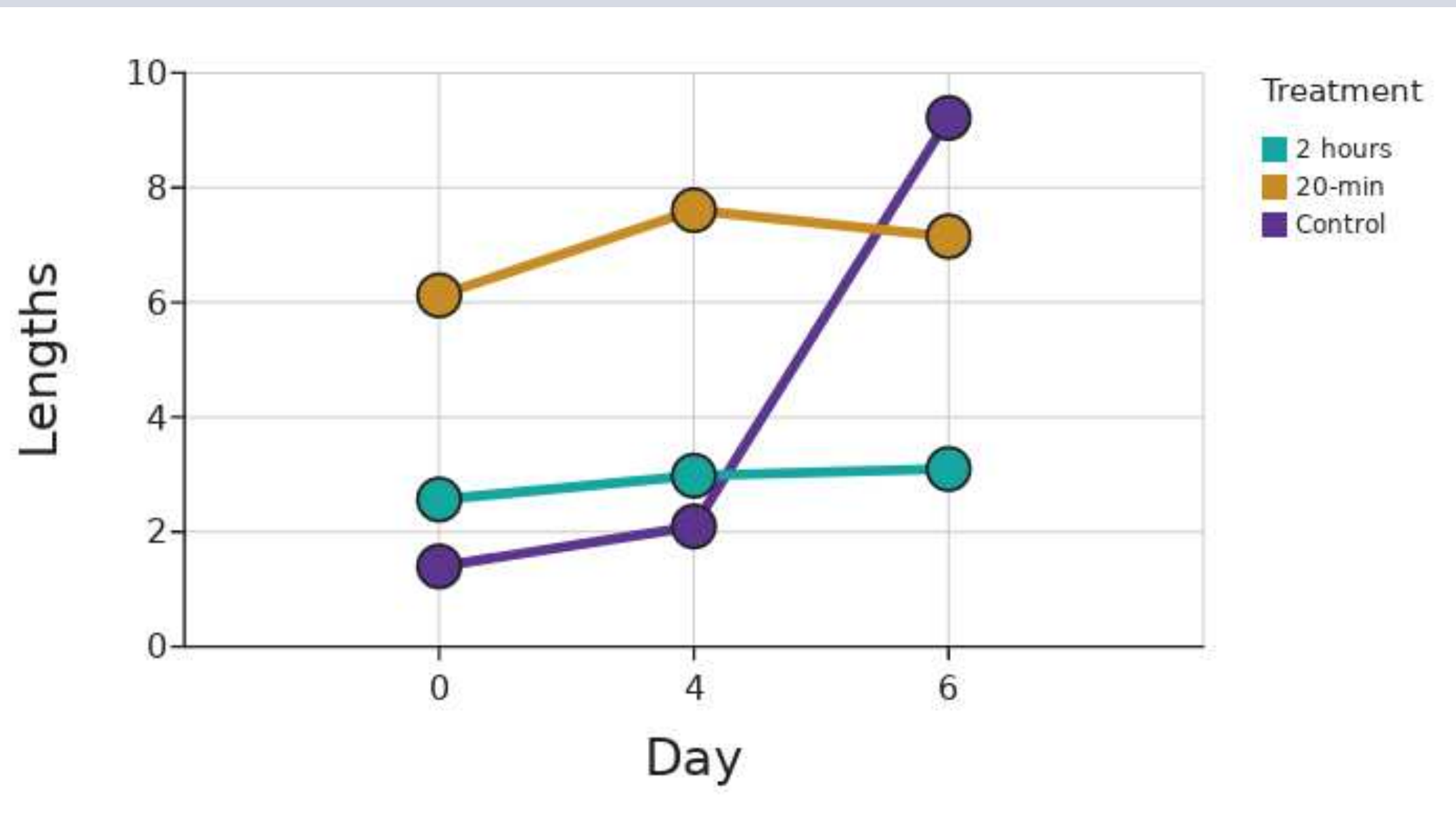


Figure 7. The graph shows the greatest amount of regeneration in the control group and least, (almost none) in the longest treatment.

# Conclusions

The results of this experiment show that increasing exposure to microplastics from nylon tea bags **negatively affects** planarian regeneration. Planarians in the control group exhibited the greatest increase in length by Day 6, indicating normal and rapid regeneration. The 20-minute exposure group showed moderate regeneration, suggesting that short-term exposure to microplastics has a limited but noticeable effect. In contrast, the 2-hour exposure group showed minimal growth, demonstrating that **prolonged exposure significantly inhibits regeneration**.

Statistical analysis further supported these observations. A Kruskal-Wallis test revealed a **statistically significant difference** between the treatment groups ( $H = 8.724$ ,  $df = 2$ ,  $p = 0.013$ ), allowing the null hypothesis to be rejected. Post-hoc Dunn's pairwise comparisons showed significant differences between the control and 20-minute exposure groups ( $p = 0.041$ ), the control and 2-hour exposure groups ( $p = 0.008$ ), and the 20-minute and 2-hour exposure groups ( $p = 0.036$ ).

These findings support the hypothesis that microplastics impair biological processes, likely by interfering with cellular activity involved in tissue regeneration. The clear decrease in regeneration as exposure time increased suggests a **dose-dependent effect** of microplastics on organism health.

To better connect these findings to human health, future studies could investigate how microplastics affect human cells, particularly those involved in tissue repair and regeneration. For example, experiments could expose cultured human cells to microplastic particles and measure changes in cell growth, gene expression, or stress responses. Additional research could also examine whether chemicals released from plastics contribute to these effects.

# References

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Methodology figures were created by student in BioRender and result graph and table was created by students in DataClassroom and Google Sheets, respectively. Unless otherwise cited, all other images were captured by student researcher.